

Technical Specification: Google Cloud Vision FaceID Solution

1. Executive Summary

The RawDrive FaceID solution is a high-performance, privacy-conscious facial recognition and organization system. It utilizes a **hybrid AI architecture** that leverages Google Cloud Vision for advanced detection and a local ArcFace model for secure biometric identity generation.

2. High-Level Architecture

The system is built as a multi-layered service within the RawDrive backend, favoring loose coupling and high availability.

Component Relationship

```
graph TD
    API["FastAPI Endpoints (/faces)"] --> DS["FaceDetectionService (Orchestrator)"]
    DS --> BCS["BiometricConsentService (GDPR Gatekeeper)"]
    DS --> PM["ProviderManager (Failover Engine)"]

    PM --> GCV["CloudVisionProvider (Primary)"]
    PM --> GEM["GeminiProvider (Fallback)"]
    PM --> LPD["LocalProvider (Offline Backup)"]

    DS --> FE["FaceEmbedder (ArcFace Identity)"]
    DS --> FCS["FaceClusterService (Grouping)"]

    FCS --> FER["FaceEmbeddingRepository (PGVector)"]
    FER --> DB["PostgreSQL + PGVector"]
```

3. Component Deep Dive

3.1 FaceDetectionService (The Orchestrator)

The master service that coordinates the entire processing pipeline.

- **Responsibilities:** Job management, workspace-level setting enforcement, and pipeline sequencing.
- **Privacy Gate:** Blocks all processing until `is_face_detection_allowed(workspace_id)` returns `True`.

3.2 CloudVisionProvider (The Detector)

Integrates with Google Cloud Vision API (`google-cloud-vision`).

- **Features Used:**
 - `FACE_DETECTION` : 30+ facial landmarks, head pose (pan, tilt, roll), and emotional likelihoods.
 - `LABEL_DETECTION` : Used for secondary tagging and scene context.
- **Failover:** If the API is unreachable, the `ProviderManager` automatically routes requests to the `GeminiProvider` or `LocalProvider`.

3.3 FaceEmbedder (The Identity Generator)

Local AI model for biometric fingerprinting.

- **Model:** [InsightFace ArcFace \(Buffalo_L\)](#).
- **Output:** 512-dimensional unit vector (embedding) per face.
- **Security:** Embeddings are generated locally using OpenCV DNN. Biometric identity data never leaves the RawDrive infrastructure.

3.4 FaceClusterService (The Organizer)

Manages the relationship between faces and identities.

- **Mechanism:** Centroid-based clustering.
- **Centroid:** The average embedding of all faces in a group, representing a person's "canonical" face.
- **Logic:** Uses cosine similarity to find the best-matching person. If similarity > 0.7, the face is added to the group and the centroid is recalculated.

4. Sequence of Operations

1. **Ingestion:** A photo is uploaded or scheduled for processing.
2. **Consent:** System verifies biometric consent.

- 3. **Detection:** `CloudVisionProvider` identifies one or more face bounding boxes and attributes.
- 4. **Identity:** For each detected face, a 112x112 crop is passed to the `FaceEmbedder`.
- 5. **Persistence:** The face record, bounding box, metadata (smile, pose), and 512-d embedding are stored in PostgreSQL.
- 6. **Grouping:** `FaceClusterService` identifies if this face belongs to an existing person or if a new group should be created.

5. Data Privacy & Compliance (GDPR)

- **Article 9 Compliance:** Biometric data is classified as "Special Category Data". RawDrive requires explicit, granular consent before enabling these features.
- **Consent Logs:** Every grant/withdrawal of consent is logged with the user ID and timestamp for audit purposes.
- **Retention:** When consent is withdrawn, users can trigger a cascade delete of all biometric embeddings and face groups.

6. Technical Specifications

Feature	Specification
Embedding Dimension	512 Floats
Similarity Metric	Cosine Similarity (via PGVector)
Primary Model	ArcFace (w600k_r50.onnx)
Vector Storage	<code>vector(512)</code> type in PostgreSQL
Detection Provider	Google Cloud Vision API v1

7. Setup & Configuration

- **Admin Settings:** Providers can be enabled/disabled per workspace.
- **Failover Threshold:** Default 5 consecutive failures triggers a temporary circuit breaker open on the provider.
- **Environment:** Requires `GOOGLE_APPLICATION_CREDENTIALS` for Cloud Vision or a service account JSON stored in the DB.